When to Update? Budget-Aware Selection of Model Updates for Clinical Prediction under Context Shift

ContextShift-Health

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Abstract

When a clinical risk model is deployed at a new site, the reflex is to retrain it. We ask a narrower question: given what changed and how few labelled outcomes the site has, *which* update is worth applying? We formalise this as an update-selection problem over a cost-ordered ladder of eight updates, from doing nothing to a full retrain, and we meta-learn a selector that reads label-free shift diagnostics plus the site's label budget. Across 8 held-out deployment sites in two open ICU datasets (720 real and 152 injected-shift deployments), we find: (i) transporting a mortality model leaves discrimination intact (mean AUROC 0.801 versus 0.773 for a locally trained model) while breaking calibration (observed-to-expected ratios from 0.53 to 2.79); (ii) which update is best depends on the label budget, with no update optimal in 29% of deployments; but (iii) our selector does *not* beat restraint. Doing nothing attains relative regret 0.0259 against the oracle, while the selector attains 0.0341. Two findings survive and are practically useful. Cross-validating on the site's own labels — the intuitive approach — harms up to 37% of sites, whereas the closed testing procedure harms few. And the two-tier analysis explains the negative result: label-free diagnostics detect covariate and observation-process shift (classifier two-sample AUC 1.00 for measurement shift) but are blind to prior and concept shift (0.49 and 0.48, both at chance) — and those are precisely the shifts that determine whether recalibration helps.

1 Introduction

A clinical prediction model does not learn about disease; it learns about a place. Trained at one hospital it absorbs that hospital's case mix, its monitoring habits, its thresholds for admission. Moved elsewhere, it keeps answering a question about the old place. This is well documented [1], and the standard response is to retrain.

Retraining is expensive and needs labelled outcomes the new site usually does not have. The literature offers a ladder of cheaper alternatives — correcting the intercept, rescaling the logit, revising a few coefficients [2, 3] — and a separate literature detects that shift has occurred [6] or estimates its effect [7]. What is missing is the decision that connects them: given this site, these diagnostics, and these few labels, which rung of the ladder should we step on?

Table 1: The update ladder, ordered by cost.

	Update	Labels
S_0	no update	0
S_1	prior (prevalence) correction, damped EM	0
S_2	intercept recalibration	≥ 1
S_3	logistic recalibration	≥ 1
S_4	importance-weighted refit	0
S_5	model revision (top-10 coefficients)	≥ 1
S_6	pooled refit, target upweighted	≥ 1
S_7	target-only retrain	≥ 1

We make three contributions. We formalise update selection with a label budget, scoring policies by regret against the update that was best in hindsight and by how often they make a site worse. We meta-learn a selector from pseudo-deployments, separating diagnostics into Tier 1 (label-free, available on day one) and Tier 2 (needs outcomes). And we report an honest negative result with a mechanism: the selector fails, and the tier analysis says why.

2 Problem formulation

A *deployment episode* consists of a source model f trained without the target site, that site's unlabelled records U_t, n labelled cases L_n drawn from its adaptation half, and a held-out evaluation half E_t that only the scorer reads. A policy π maps (f, U_t, L_n) to one strategy from the ladder $S_0 \dots S_7$ (Table 1). With Brier score as the loss,

$$\text{Regret}(\pi) = \text{Brier}_{E_t}(S_\pi) - \min_k \text{Brier}_{E_t}(S_k),$$

normalised by $\text{Brier}_{E_t}(S_0)$ so that sites of very different prevalence are comparable. A policy *harms* a site when it exceeds the no-update Brier by more than 0.002.

3 Method

Tier 1 diagnostics are computable the day the model arrives: a cross-fitted classifier two-sample test overall and per feature group (demographics, physiology, observation process, treatment), the effective sample size of importance weights, a damped EM estimate of prevalence change, and shifts in the predicted-risk distribution. **Tier 2 diagnostics** need outcomes: observed-to-expected ratio and its z-score, calibration slope and its standard error, AUROC, and a Brier "surprise" term. Each Tier 2 feature carries an availability flag, because they switch on at different sample sizes. The label budget itself is Tier 0: it is the constraint, not a diagnostic.

The selector fits one gradient-boosted regressor per strategy, predicting the relative Brier change against no update, and acts only when a predicted improvement clears a margin τ fitted on training domains. Without that margin an unguarded arg-min chases prediction noise. We also fit a depth-3 tree (SEL-RULE) whose policy can be printed.

Baselines. Every fixed policy; TCV, which cross-validates the candidate updates on the site's own labels; and CTP, the closed testing procedure of Vergouwe et al. [3] generalised to any base model. Both baselines are Tier 2: they see outcomes only.

4 Experimental setup

Two open-access datasets, no credentialing required. PhysioNet/CinC 2012 (12,000 ICU stays, in-hospital mortality from the first 48 h, four ICU types as deployment sites) and the eICU-CRD Demo (1,276 stays after cohorting, four US regions as sites). Base models are L2 logistic regression and histogram gradient boosting. Source models are trained leave-one-site-out; each site is split 50/50 into an adaptation half and an evaluation half. Budgets are $n \in \{0, 10, 25, 50, 100, 200, 400, 800\}$, capped by site size, with 8 repeats. Selectors are trained on the other sites plus 152 injected-shift pseudo-deployments (covariate, prior, observation and concept shift at three severities) and tested only on real held-out sites. Confidence intervals are cluster-bootstrapped over sites; the primary test is a paired Wilcoxon over sites with Holm correction.

5 Results

5.1 Discrimination transfers; calibration does not (H1)

Figure 1 shows the pattern at every site. A transported model ranks patients as well as a locally trained one (mean AUROC 0.801 versus 0.773), while its probabilities drift: the observed-to-expected ratio ranges from 0.53 to 2.79 and the calibration slope falls as low as 0.39.

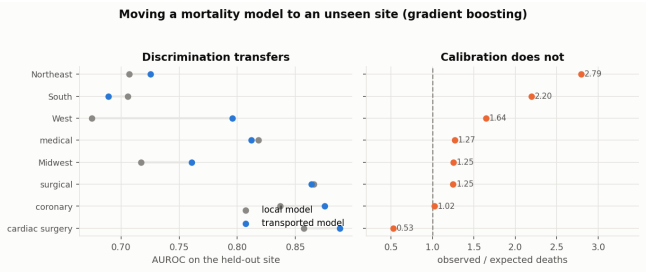


Figure 1: Each site, gradient boosting. Ranking survives the move; calibration does not.

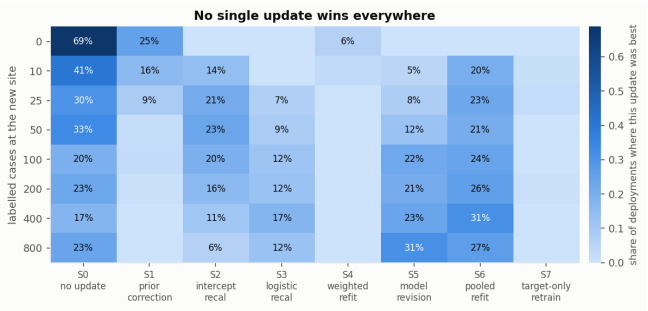


Figure 2: Share of deployments in which each update was best in hindsight.

Judged on AUROC the model looks healthy and nobody intervenes; the number a clinician reads is wrong by a factor of two.

5.2 The best update depends on the budget (H2)

Figure 2 maps which update was best in hindsight. With no labels, doing nothing wins 69% of the time; by 800 labels the flexible updates (revision, pooled refit) win most deployments. No single strategy is best in more than a third of episodes overall, and the available gain is small: the oracle beats no update by 1.90×10^{-3} Brier on average.

5.3 The selector does not beat restraint (H3 rejected)

Table 2 is the central result and it is negative. In the low-budget regime, doing nothing attains relative regret 0.0259; our selector attains 0.0341 and the closed testing procedure 0.0269. Against the pre-registered criterion — a 20% regret reduction with the confidence interval excluding zero — the selector fails against all three competitors (Holm-adjusted p from 0.094, all in the wrong direction except against TCV).

The selector is also unsafe: it harms 14% of low-budget deployments against a 5% target (H4 rejected). Its failure mode is visible per site: it acts in every episode at one eICU region, costing 3.3×10^{-3} Brier there, while

Table 2: Low-budget regime ($n \leq 100$), leave-one-site-out. Relative regret against the oracle (lower better), share of sites harmed, and the mean cost rank of the chosen update.

Policy	Rel. regret	Harm	Cost
oracle (unattainable)	0.000	0.00	2.6
no update	0.0259	0.00	0.0
CTP [3]	0.0269	0.02	0.2
SEL-META (both tiers)	0.0341	0.14	0.8
SEL-META (Tier 1 only)	0.0349	0.14	1.1
SEL-RULE	0.0389	0.19	2.7
TCV	0.0591	0.25	1.9
always recalibrate intercept	0.0440	0.25	2.0
always retrain on target	0.3607	0.81	7.0

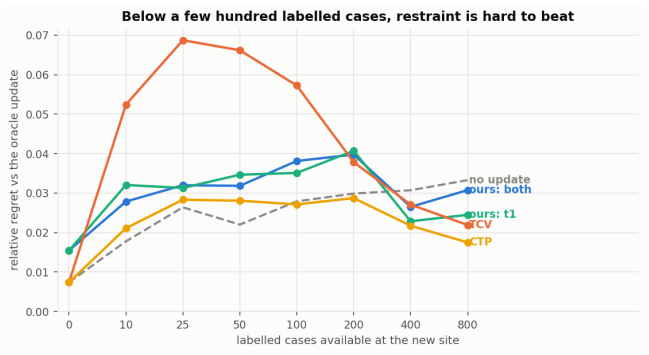


Figure 3: Relative regret against label budget. Restraint (dashed) is hard to beat until several hundred labelled outcomes are available.

acting correctly at the badly calibrated cardiac-surgery unit.

5.4 What does work

Two positive findings survive. First, above roughly 400 labelled outcomes the picture inverts and updating pays: the best policy in that regime is S2 with relative regret 0.0157, against 0.0320 for doing nothing. Second, CTP is the sound default in the low-budget regime: near-zero cost, 2% of sites harmed, and it acts only when a likelihood-ratio test says the update is warranted. By contrast TCV — cross-validating on the site's own 25 to 100 labels, which is what a careful practitioner would naturally do — harms up to 37% of sites. Small-sample validation is worse than no validation.

5.5 Why the selector fails (H7 and the tier analysis)

The injected-shift experiments give the mechanism. Label-free diagnostics see what moves in feature space and nothing else: measurement shift is detected almost perfectly (C2ST AUC 1.00, and the observation-process feature group ranks first), covariate shift is detected

Table 3: Injected shifts: which feature group the diagnostics flag, at $n = 0$.

injected shift	G1	G2	G3	G4
concept	4	0	0	2
covariate	3	2	0	1
none	2	0	0	0
observation	0	2	4	0
prior	2	2	1	1

weakly (0.60), while prior shift (0.49) and concept shift (0.48) sit at chance, as they must: neither changes the marginal distribution of X. Under no shift the test correctly reports 0.47.

But prior and concept shift are exactly what decide whether recalibration helps. So Tier 1, the abundant information, is structurally blind to the question the selector must answer, and Tier 2, which can see it, is too noisy at 25–100 labels to answer it reliably. Permutation importance confirms this: the label budget dominates (Tier 0 total 0.27, Tier 1 0.05, Tier 2 -0.13, all $\times 10^{-3}$ Brier). The selector largely learned how many labels it had, not what had changed.

6 Discussion

The finding that updating rarely helps at small n is not an artifact of weak updates: the same updates help at the same sites once several hundred labels are available, and the intercept correction demonstrably fixes calibration where calibration is broken (at the cardiac-surgery unit it moves the observed-to-expected ratio from 0.46 to 0.54 and ECE from 0.056 to 0.041). The problem is that at 25–100 labels the evidence for acting is indistinguishable from noise, and acting on noise costs more than it gains.

For practice this suggests a simple and slightly deflating rule: below roughly a hundred outcomes, leave the model alone and monitor; use a closed testing procedure if you must act; do not cross-validate your way to a decision on a handful of cases; and once a few hundred outcomes exist, recalibrate the intercept.

7 Limitations

The deployment sites are ICU types and US regions rather than individual hospitals, because the open eICU demo spreads 2,520 stays over 186 hospitals; with eight sites, confidence intervals are wide and the primary test has little power. The oracle is computed on finite evaluation halves, so part of the oracle gain is noise a real selector cannot capture, which inflates every regret. The 2012 dataset is single-centre, so its sites differ by case mix rather than by institution. Everything here should

be re-run on full eICU-CRD (208 hospitals) and MIMIC-IV, for which the code is unchanged.

8 Conclusion

Choosing how much to adapt a clinical model can be posed cleanly as an update selection problem, and the phase diagram of which update wins where is useful on its own. But the central hypothesis — that label-free shift diagnostics can guide that choice when labels are scarce — does not survive contact with the data, and the reason is structural: the shifts that decide whether an update helps are invisible without labels. Restraint remains the strong baseline.

References

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